CRYPTOGRAPHY AND NETWORK SECURITY	
Course Code: CY53	**Credits: 4:0:0**
Prerequisite: Nil	**Contact Hours: 56L**
Course Coordinator: Mrs. Shankaramma	

Course Contents

Unit I

Overview: Computer Security Concepts, The OSI Security Architecture, Security Attacks, Security Services, Security Mechanisms, A Model for Network Security. **Classical Encryption Techniques**: Symmetric Cipher Model: Cryptography, Cryptanalysis and BruteForce Attack, Substitution Techniques: Caesar Cipher, Mono-alphabetic Cipher, Playfair Cipher, Poly alphabetic Cipher..

- Pedagogy: Chalk and Talk, PowerPoint Presentations

Unit II

Block Ciphers and the data encryption standard: Stream Ciphers and Block Ciphers, the Feistel Cipher, DES encryption, DES decryption. A DES example: Results, The Avalanche effect, The Strength of DES: The Use of 56-Bit Keys, the nature of the DES algorithm, timing attacks, Block Cipher Design Principles: Number of rounds, Design of function F, Key schedule Algorithm. **Advanced Encryption Standard**: AES structure, AES Transformation Functions, AES Key Expansion.

- Pedagogy: Chalk and Talk, PowerPoint Presentations

Unit III

Public-Key Cryptography and RSA: Principles of public-key cryptosystems: Public-key cryptosystems. Applications for public-key cryptosystems, Requirements for public-key cryptosystems. Public-key cryptanalysis. **The RSA algorithm**: Description of the algorithm, computational aspects, the security of RSA. **Other Public Key Cryptosystems**: Diffie Hellman Exchange: The Algorithm, Key Exchange Protocols, Man in the middle attack.

- Pedagogy: Chalk and Talk, PowerPoint Presentations, Active Learning

Unit IV

Key Management and Distribution: Distribution of Public Keys: Public Announcements of Public Keys, Public Available Directory, Public Key Authority, Public Key Certificates, X-509 certificates. Certificates, X-509 version 3, Kerberos, Kerberos version 4. **Web Security Considerations**: Web Security Threats, Web Traffic Security Approaches.

- Pedagogy: Chalk and Talk, PowerPoint Presentations.

Unit V

Secure Sockets Layer: SSL Architecture, SSL Record Protocol, Change Cipher Spec Protocol, Alert Protocol, and Handshake Protocol, Cryptographic Computations. **Transport Layer Security**: Version Number, Message Authentication Code, Pseudorandom Functions, Alert Codes, Cipher Suites, Client Certificate Types, Certificate Verify and Finished Messages, Cryptographic Computations, and Padding. HTTPS Connection Initiation, Connection Closure. Secure Shell(SSH) Transport Layer Protocol, User Authentication Protocol, Connection Protocol. **Electronic Mail**

Security: Pretty Good privacy, S/MIME.

- Pedagogy: Chalk and Talk, PowerPoint Presentations

Text Books:

1. William Stallings, “Cryptography and Network Security”, Pearson Education, 6th ed, 2013.

References:

1. William Stallings, “Network Security Essentials Applications and Standards”, 2nd ed., Pearson Education, 2003.
2. Charlie Kaufman, Radis Perlman and Mike Speciner, “Network Security – Private Communication in a Public World” 2nd ed., Pearson Education, 2003.
3. Cyrus Piekari, Anton Chuvakin, “Security Warrior”, 2nd ed., Oreilly Publishers, 2005.
4. Peborab Russell, G.T. Gangeni Sr, “Computer Security Basics”, 2 nd ed., O’reilly Publishers, 2006.

Course Outcomes (COs):

At the end of the course the student will be able to:

1. Describe computer security concepts,security attacks, services, mechanisms, and the OSI security architecture. (PO 1,2 ,PSO-1)
2. Apply classical encryption techniques, symmetric encryption algorithms such as DES and AES. (PO 1,2,3,5 , PSO-1,2,3)
3. Implement public-key cryptography algorithms like RSA and Diffie-Hellman, and analyse their security. (PO 1,2,3,4,5, PSO-1,2,3)
4. Demonstrate key management techniques and user authentication protocols,Kerberos and digital certificates. (PO 1,2,3, 5 PSO-1,2,3)
5. Apply security protocols such as SSL/TLS, SSH, PGP, and S/MIME for securing web applications. (PO 1,2,3,5 ,PSO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	**Marks**	**Course Outcomes (COs) addressed**
Internal Test-I (CIE-I)	30	CO1, CO2
Internal Test-II CIE-II)	30	CO3, CO4, CO5
Average of the two CIE shall be taken for 30 marks		
Other Components		
Assignment-1	10	CO1, CO2, CO3
Assignment-2	10	CO4, CO5
The Final CIE out of 50 Marks = Average of two CIE tests for 30 Marks+ Marks scored in Assignment-1 +Marks scored in Assignment-2		
Semester End Examination (SEE)	50	CO1, CO2, CO3 CO4, CO5